

ME-203A: Kinematics of Machines

by

Dr. Emarti Kumari



**Department of Mechanical Engineering
M.B.M. Engineering College
Jai Narain Vyas University, Jodhpur**

UNIT 1

Kinematic analysis of mechanism: Definitions of Links, pairs and kinematic chain, Mechanism and machine, Inversion of four bar kinematic chain and slider crank chains.

Velocity and acceleration analysis: Relative velocity method; Instantaneous center method, Coriolis component of acceleration

➤ Degree of freedom (DOF):

- ❖ An constrained rigid body moving in space can have three translations (u, v, w) and three rotational ($\theta_x, \theta_y, \theta_z$) motions about the three mutually perpendicular axes.
- ❖ Degree of freedom of a kinematic pair is defined as the number of independent relative motions, both translational and rotational.
- ❖ **Minimum number of independent coordinates (variables)** that need to be specified to fix the configuration of a mechanism.
- ❖ One link of the chain is grounded.

Degree of freedom = 6 – Number of restraints

The degree of freedom of some of systems are:

- Rigid body has six degree of freedom
- Rectangular bar sliding in a rectangular hole has one degree of freedom
- The Piston of the crank of a slider crank mechanism can be expressed by angular motion (one degree of freedom)
- A circular shaft rotating in a hole and also transmitting parallel to its axis has two degree of freedom
- A Ball and a socket joint has three degree of freedom

- Number of links = n_L
- Number of joints = n_J
- Degree of freedom of i^{th} joint = f_i

Degree of freedom of **planar mechanisms**:

$$F = 3(n_L - 1) - 3n_J + \sum_{i=1}^{n_J} f_i$$

Degree of freedom of **Spatial mechanisms**:

$$F = 6(n_L - 1) - 6n_J + \sum_{i=1}^{n_J} f_i$$

Degree of freedom (DOF)

- ❖ A mechanism must have at least 1 degree of freedom.
- ❖ A structure has zero degree of freedom.
- ❖ An over-constrained structure has negative degree of freedom.

Kutzbach criterion: The Kutbach criterion for determining the number of degree of freedom of a planar mechanism is:

$$F = 3(n - 1) - 2p - h$$

Here, F = degree of freedom

n = total number of links in a mechanism out of which one is fixed link

$n - 1$ = number of movable links

p = number of simple joints or lower pairs having one degree of freedom.

h = number of higher pairs having two degree of freedom or three degree of freedom

Gruebler's criterion: The Gruebler's criterion for determining the number of degree of freedom of planar mechanism with lower pairs only:

$$F = 3(n - 1) - 2p$$

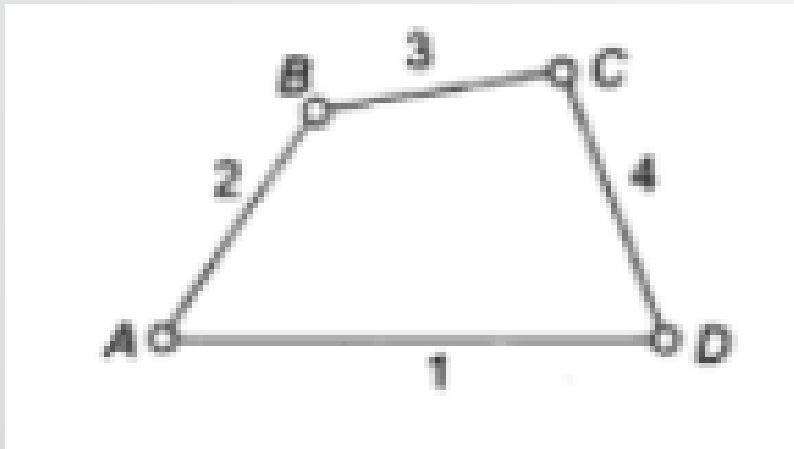
Here, F = degree of freedom

n = total number of links in a mechanism out of which one is fixed link

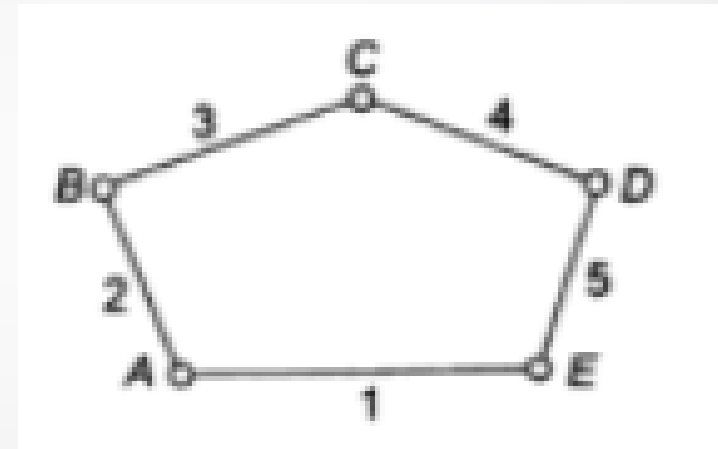
$n - 1$ = number of movable links

p = number of simple joints or lower pairs having one degree of freedom.

Example 1. Calculate the number of degree of freedom of the linkage shown in Fig.



(a)



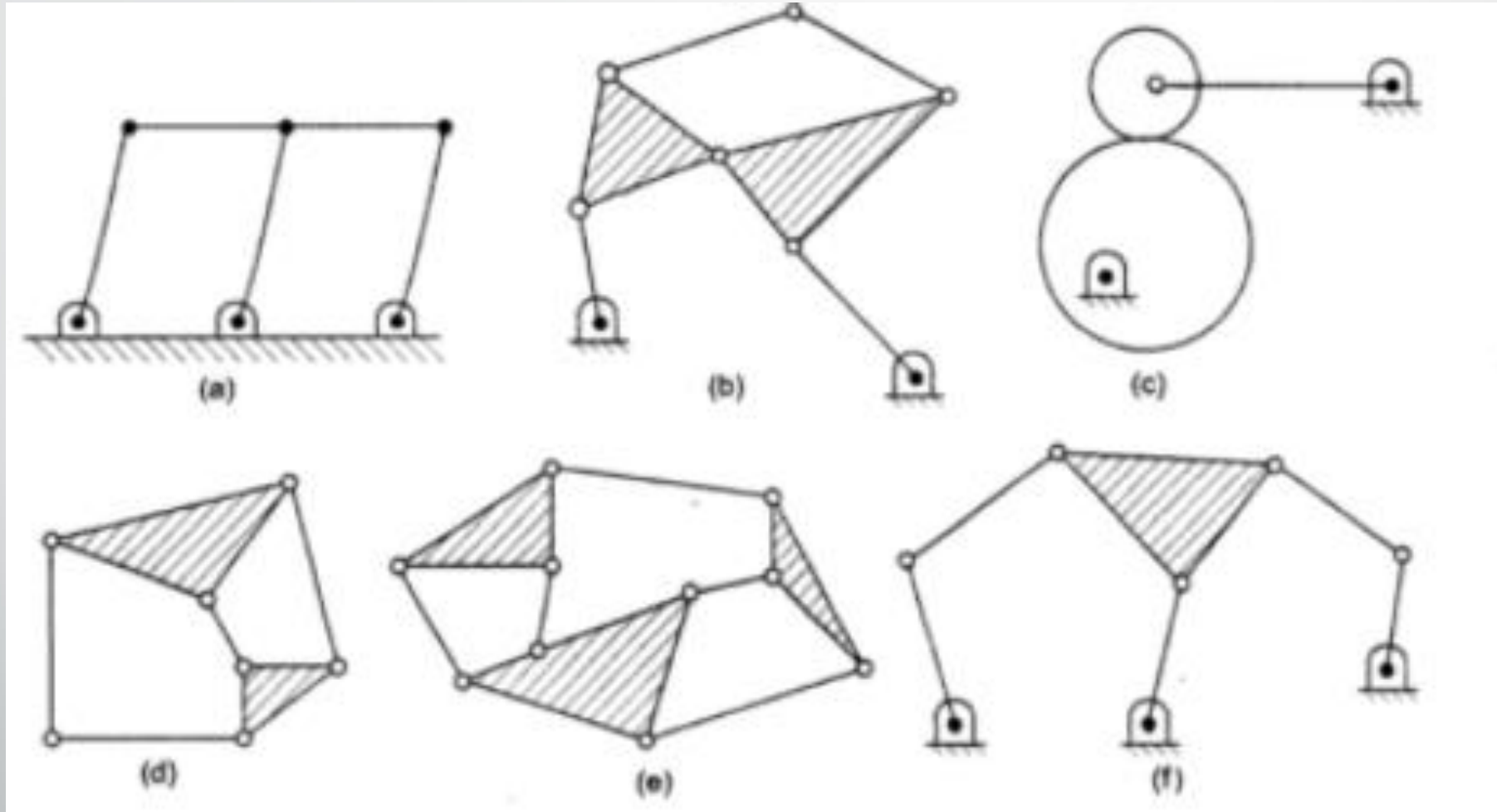
(b)

Solution:

(a) Number of binary links $n = 4$
Number of lower pairs $p = 4$
Degree of freedom $F = 3(n-1) - 2p$
 $= 3(4-1) - 2 \times 4$
 $= 1$

(b) Number of binary links $n = 5$
Number of lower pairs $p = 5$
Degree of freedom $F = 3(n-1) - 2p$
 $= 3(5-1) - 2 \times 5$
 $= 2$

Example 2. Calculate the number of degree of freedom of the linkage shown in Fig.



Solution:

(a) Number of binary links $n_2 = 3$
Number of ternary links $n_3 = 2$
Total number of links $n = n_2 + n_3 = 5$
Number of lower pairs $2p = 2n_2 + 3n_3$
$$= 2 \times 3 + 3 \times 2$$
$$= 12$$

Degree of freedom $F = 3(n-1) - 2p$
$$= 3(5-1) - 12$$
$$= 0$$

(b) Number of binary links $n_2 = 5$
Number of ternary links $n_3 = 2$
Total number of links $n = n_2 + n_3 = 7$
Number of lower pairs $2p = 2n_2 + 3n_3$
$$= 2 \times 5 + 3 \times 2$$
$$= 16$$

Degree of freedom $F = 3(n-1) - 2p$
$$= 3(7-1) - 16$$
$$= 2$$

Solution:

(c) Number of binary links $n_2 = 3$

Number of ternary links $n_3 = 0$

Total number of links $n = n_2 + n_3 = 5$

Number of lower pairs $p = 2$

Number of higher pairs $h = 1$

Degree of freedom $F = 3(n-1) - 2p - h$
 $= 3(5-1) - 4 - 1$
 $= 1$

(d) Number of binary links $n_2 = 4$

Number of ternary links $n_3 = 2$

Total number of links $n = n_2 + n_3 = 6$

Number of lower pairs $2p = 2n_2 + 3n_3$
 $= 2 \times 4 + 3 \times 2$
 $= 14$

Degree of freedom $F = 3(n-1) - 2p$
 $= 3(6-1) - 14$
 $= 1$

Solution:

(e) Number of binary links $n_2 = 5$

Number of ternary links $n_3 = 2$

Number of quaternary links $n_4 = 1$

Total number of links $n = n_2 + n_3 + n_4 = 8$

Number of lower pairs $2p = 2n_2 + 3n_3 + 4n_4$
 $= 2 \times 5 + 3 \times 2 + 4 \times 1$
 $= 20$

Degree of freedom $F = 3(n-1) - 2p$
 $= 3(8-1) - 20$
 $= 1$

(f) Number of binary links $n_2 = 5$

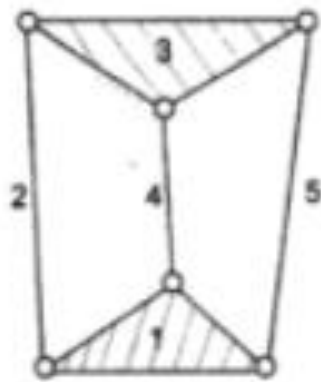
Number of ternary links $n_3 = 2$

Total number of links $n = n_2 + n_3 = 7$

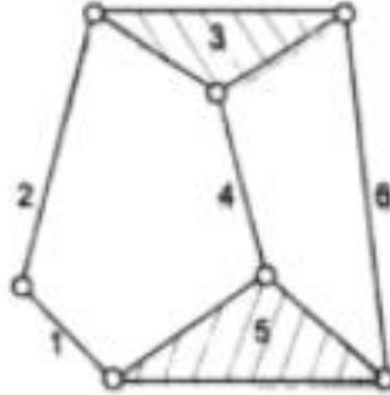
Number of lower pairs $2p = 2n_2 + 3n_3$
 $= 2 \times 5 + 3 \times 2$
 $= 16$

Degree of freedom $F = 3(n-1) - 2p$
 $= 3(7-1) - 16$
 $= 2$

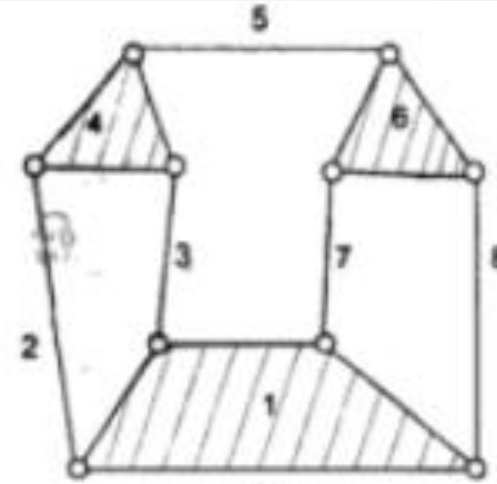
Example 3. Calculate the number of degree of freedom of the linkage shown in Fig. (for practice).



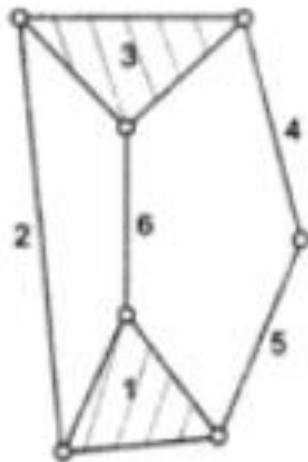
(a)



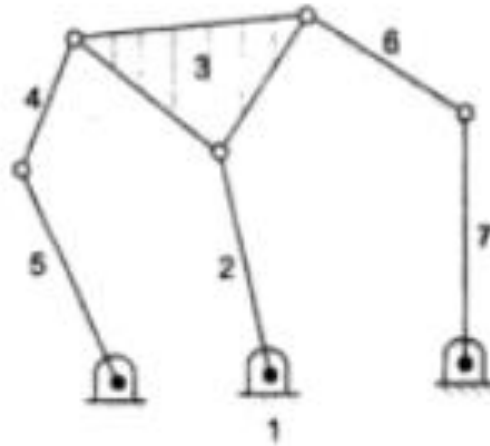
(b)



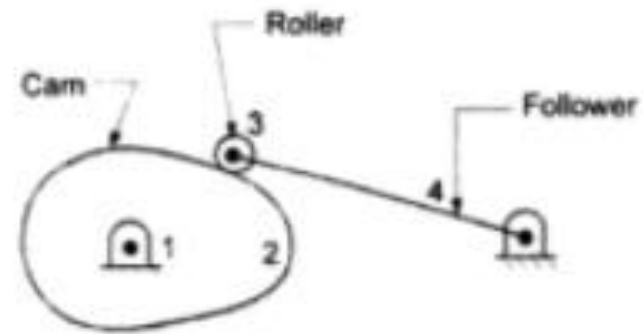
(c)



(d)



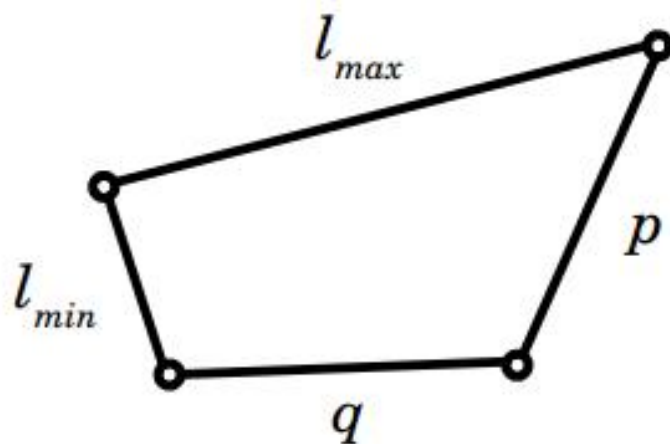
(e)



(f)

Grashof's Law or Grashof criterion (4R Chain):

This law states that for a four-bar mechanism the sum of the length of the largest and the shortest links should be less than or equal to the sum of the lengths of the other links.



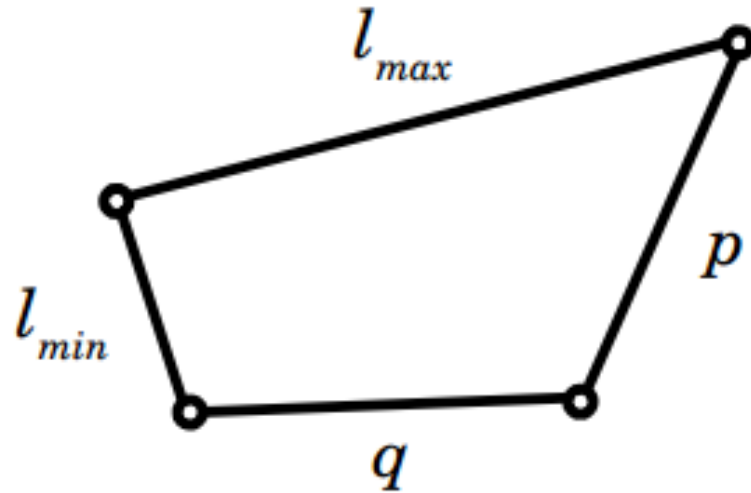
- l_{min} : length of shortest link
- l_{max} : length of longest link
- p and q : length of other two links

If

$$l_{min} + l_{max} \leq p + q$$

then the shortest link can rotate completely with respect to all other links

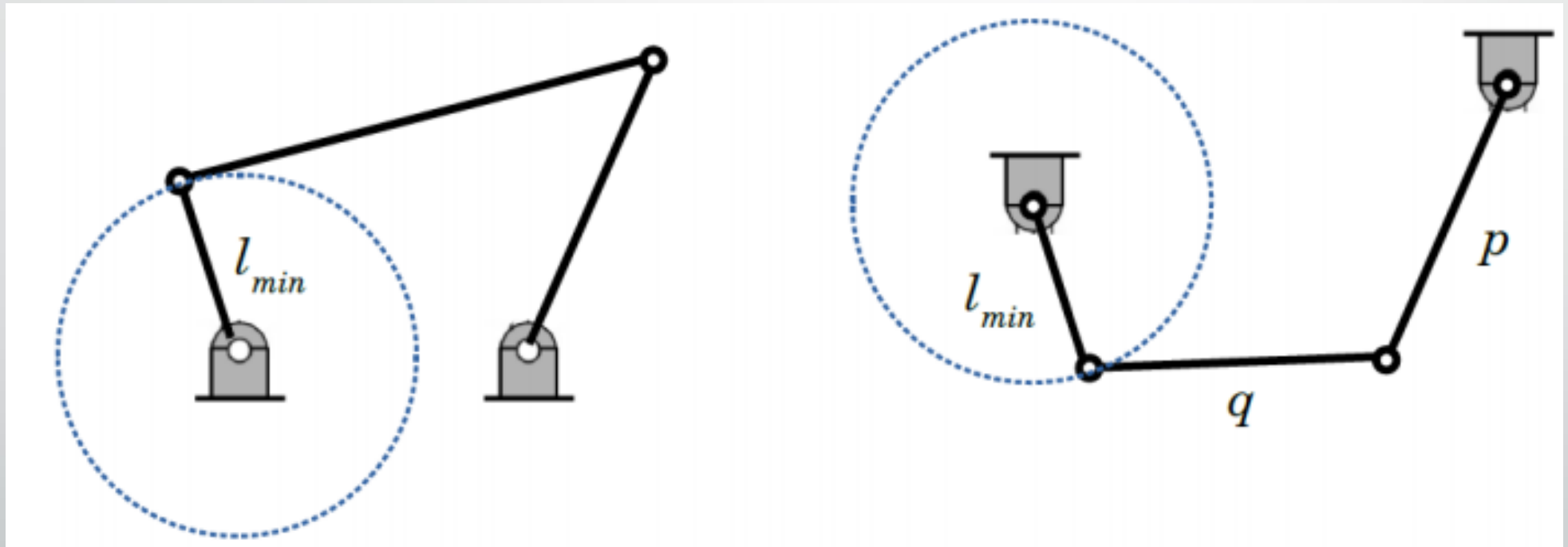
Grashof 4R chain



- Shortest link rotates with respect to all other links
- All other links rock with respect to each other

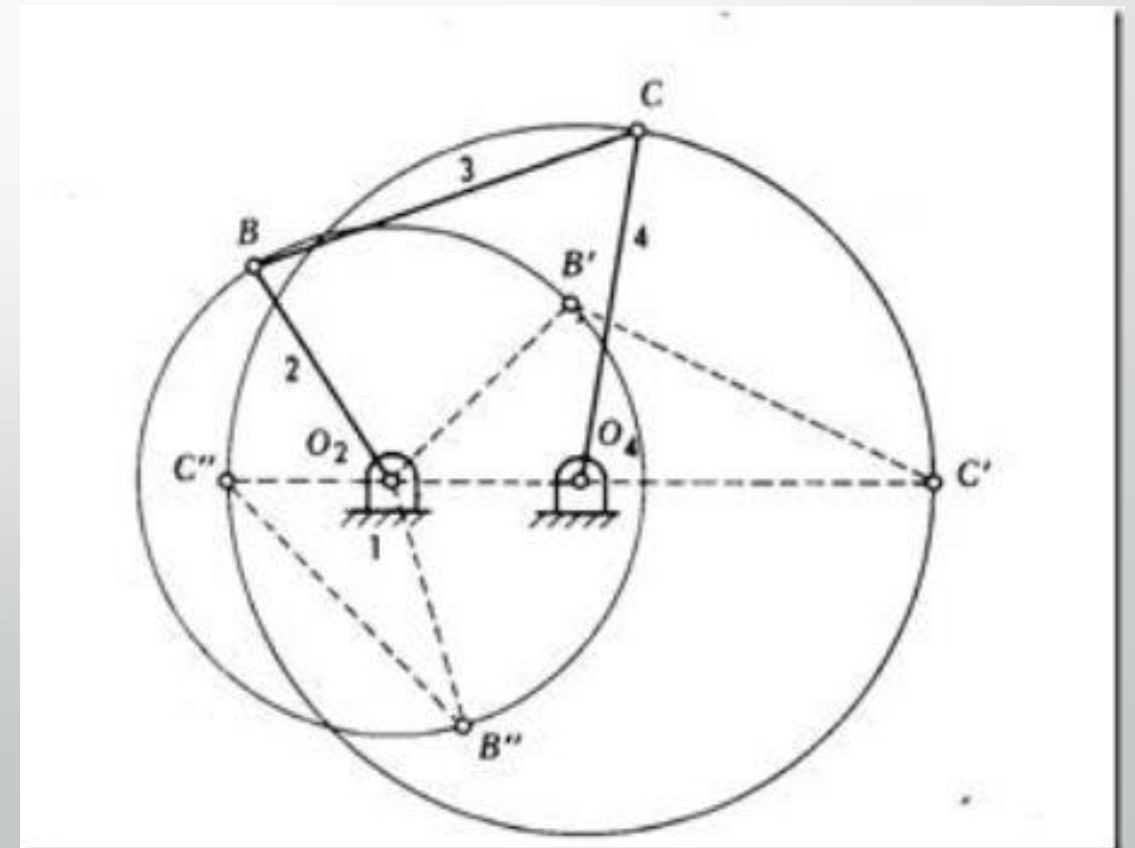
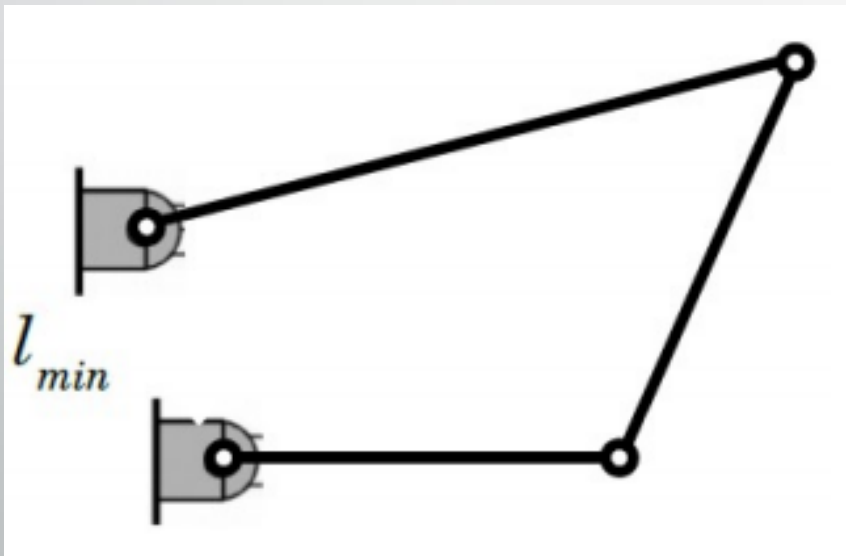
Inversions of for-bar chain: various possibilities depending on choice of ground link.

1. Crank-rocker mechanism



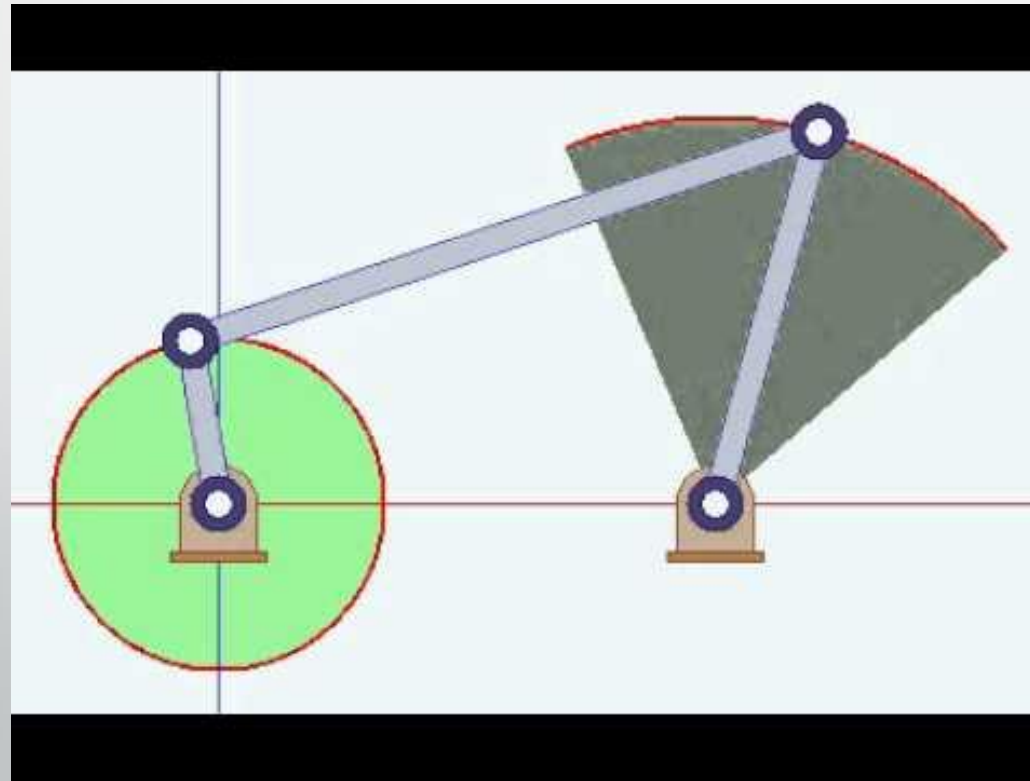
Inversions of for-bar chain: various possibilities depending on choice of ground link.

2. Crank-Crank (or double crank) mechanism: when the shortest link is fixed and $l_{min} + l_{max} \leq p + q$, crank-crank or double crank mechanism is obtained.



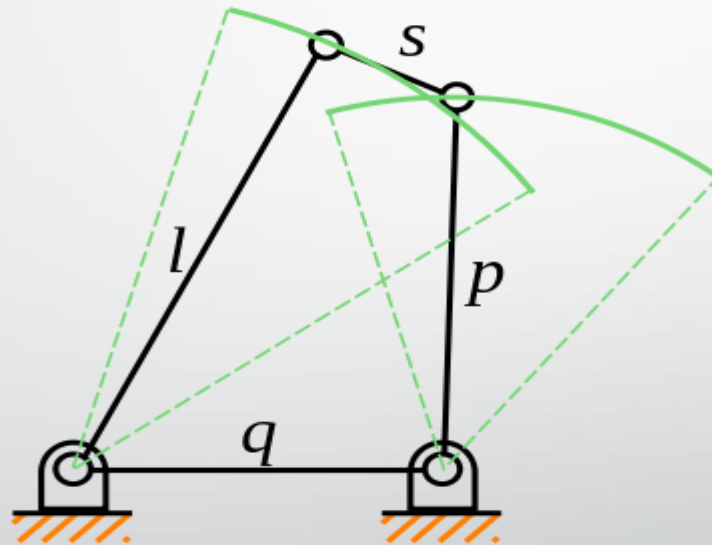
Inversions of for-bar chain: various possibilities depending on choice of ground link.

3. Crank-Rocker (or lever) mechanism: if any of the adjacent link of shortest link is fixed than shortest link have full revolution and the link L3 opposite to shortest link can oscillates.

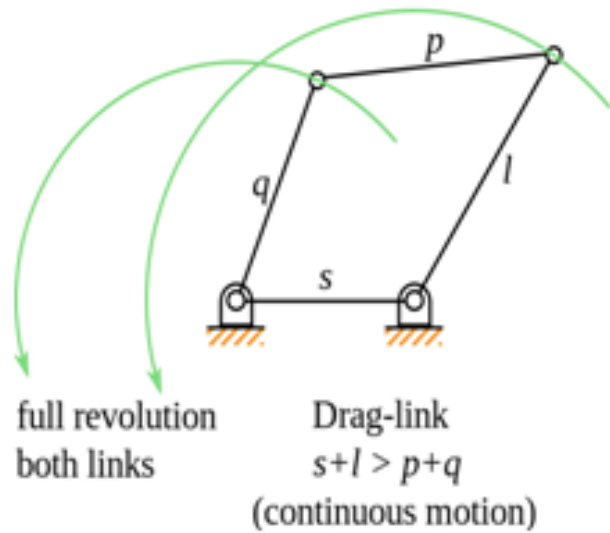


Inversions of for-bar chain: various possibilities depending on choice of ground link.

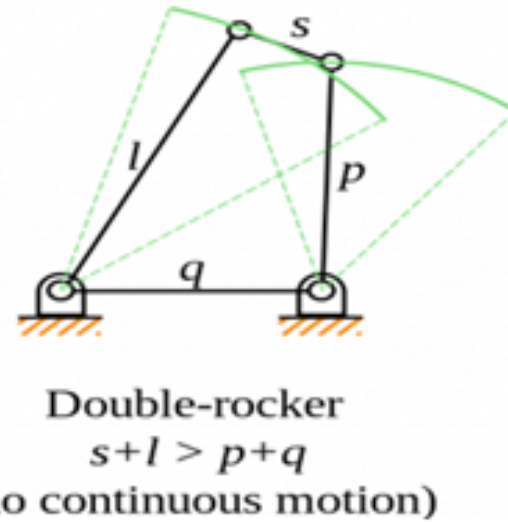
4. Rocker-Rocker (or double rocker) mechanism: if link L3 opposite to shortest link L1 is fixed and shortest link L1 is made the coupler, the other two links L2 and L4 would oscillate.



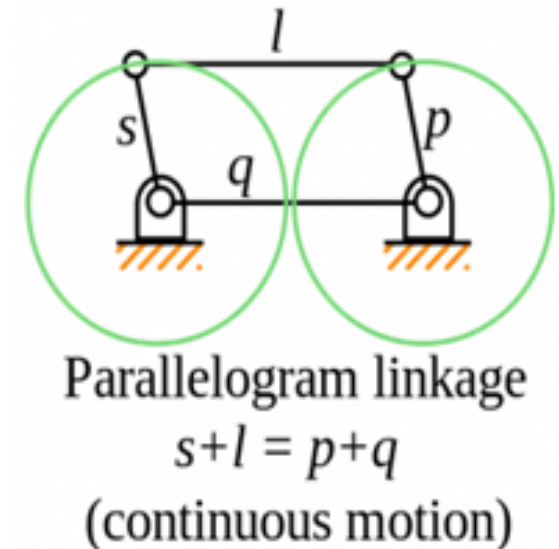
Double-rocker
 $s+l > p+q$
(no continuous motion)



(a) Double crank mechanism



(b) Double rocker mechanism



(c) Parallelogram linkage

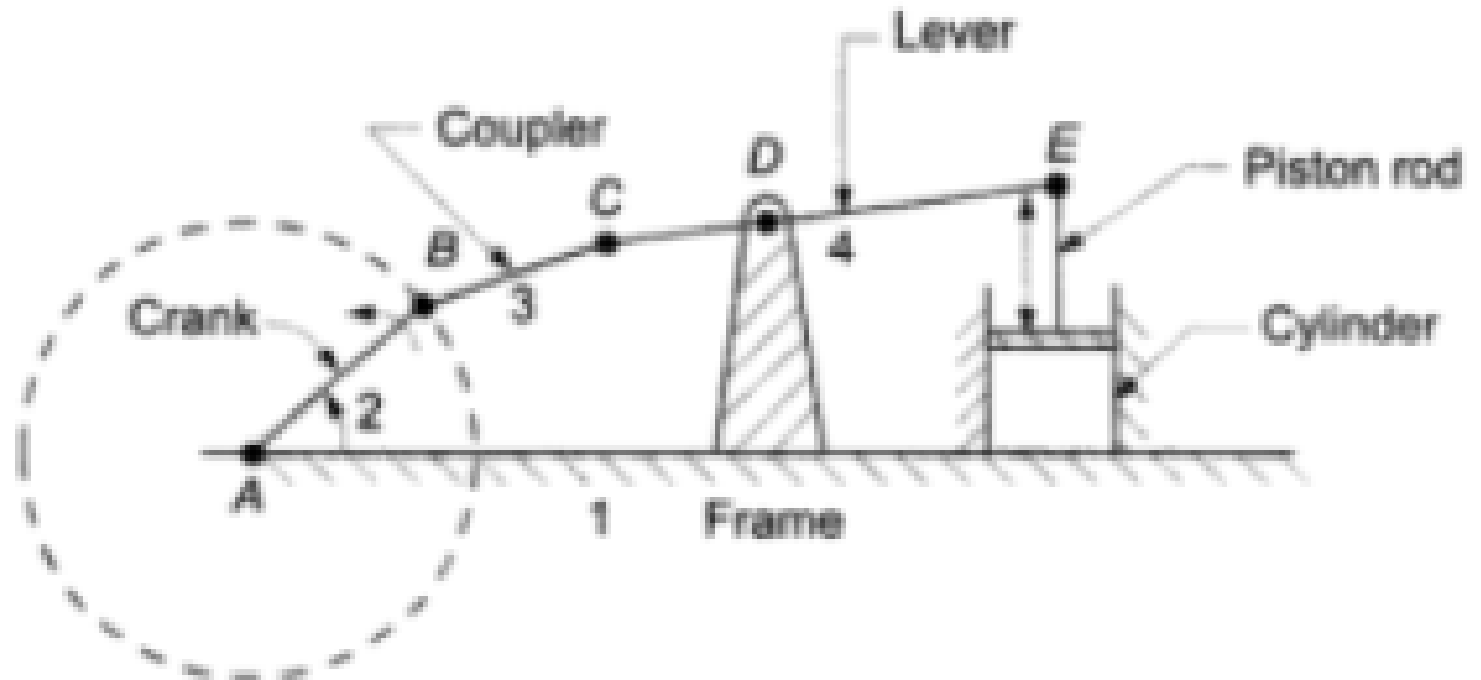
Fig. 4 Possible mechanism according to Grashof's law

Inversions of Mechanisms:

- A kinematic chain becomes a mechanism when one of the link is fixed.
- Therefore, as many number of mechanisms can be obtained as many are the links in the kinematic chain.
- This method of obtaining different mechanisms by fixing different links of a kinematic chain is called inversion of the mechanism.
- The relative motion between the various links is not altered as a result of inversion, but their absolute motion with respect to the fixed link may alter drastically.

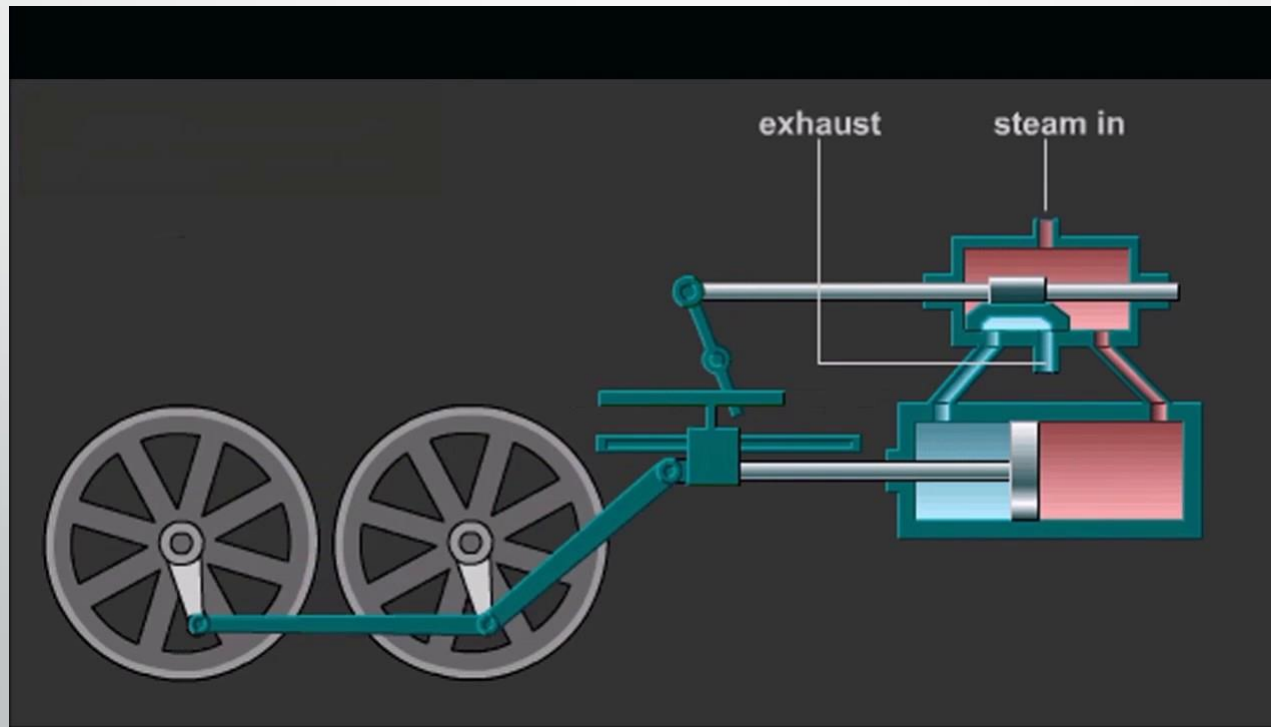
Inversions of a for-bar chain are:

- Beam engine (crank and lever mechanism): a beam engine is a type of steam engine where a pivoted overhead beam is used to apply the force from a vertical piston to a vertical connecting rod.
- Beam engines were first used to pump water out of mines or into canals.



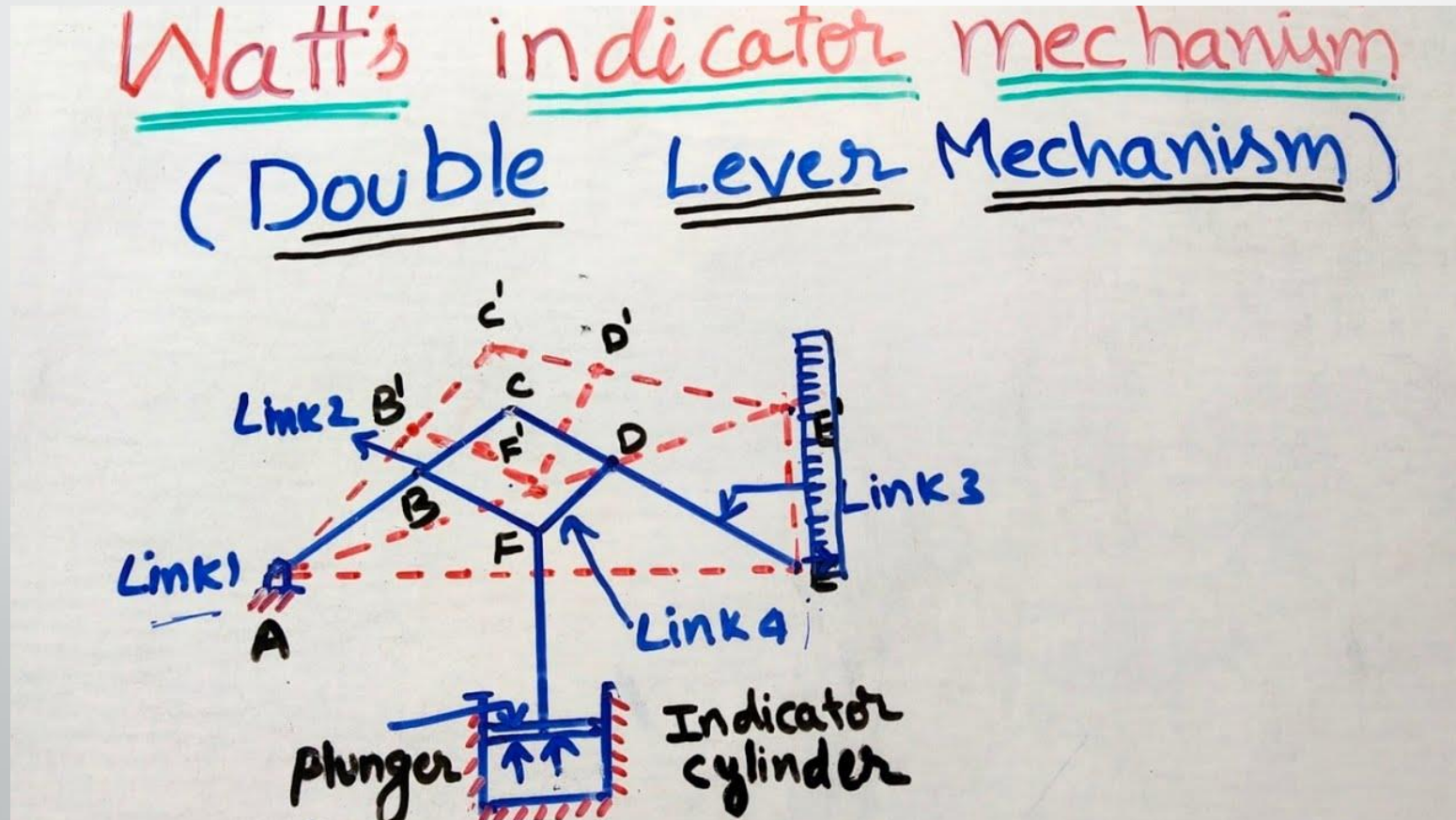
Coupled wheel of locomotive:

- The links AB and CD are of equal lengths and act as crank.
- These cranks are connected to the respective wheels.
- This mechanism is used to transmit rotary motion from one wheel to the other.
- This is also called the **double crank mechanism**.



Inversions of a for-bar chain are:

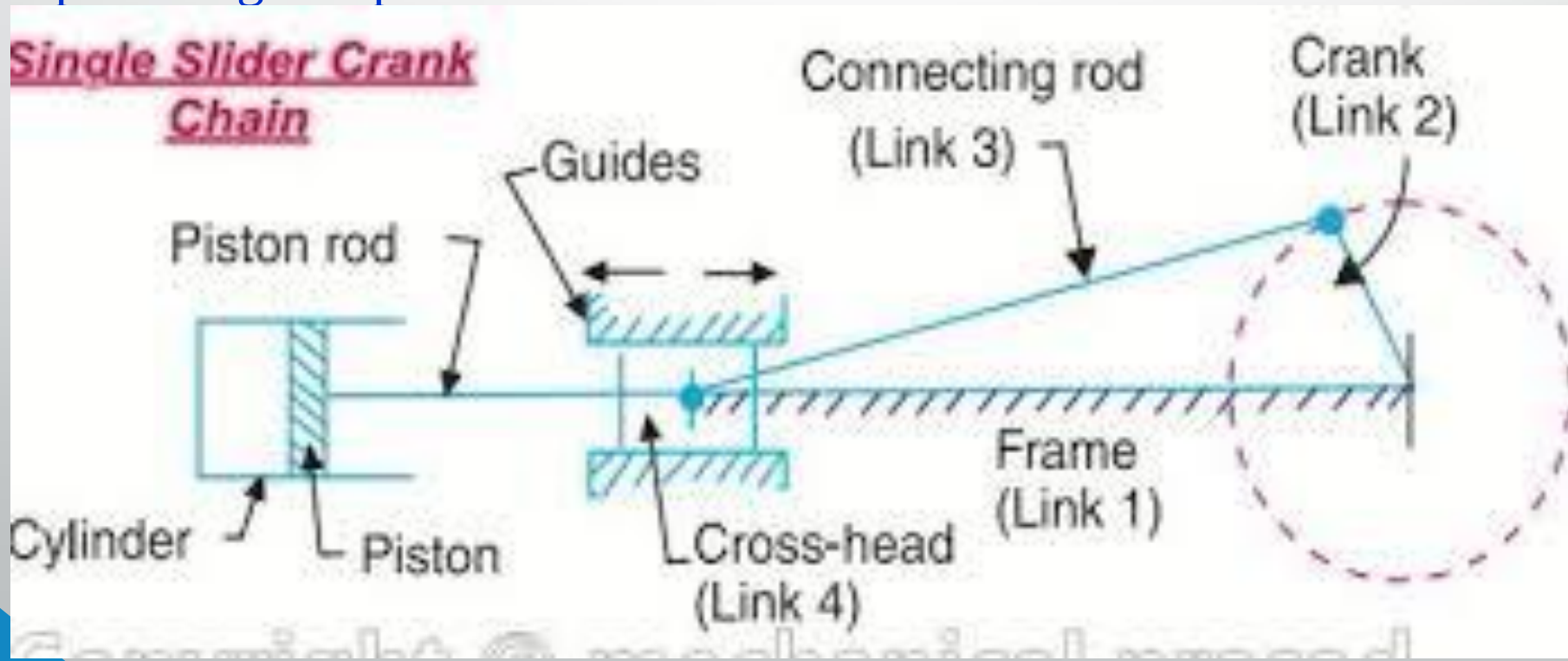
- Watt's indicator mechanism: it consists of four links: a fixed link at A, link AC, link CE, and link BFD.
- The links CE and BFD acts as levers.
- The displacement of link BFD is directly proportional to the pressure in the indicator cylinder.



Inversions of a for-bar chain are:

Single slider-crank chain: it consists of three turning pairs and one sliding pair.

- It is used to convert rotary motion to reciprocating and vice-versa.
- Applications: steam engine, internal combustion engine, reciprocating compressor.





Thank You